CSSE 220 – Object-Oriented Software Development
Rose-Hulman Institute of Technology

Worksheet 10

Name (Print):_____ Section:_____

Access & Scope

1. **Null vs empty**

```
int[][] a = null;

int[][] a = new int[0][0];
```

null means: _____
empty array means: _____

2. **Complete checks**

Type	Check for null	Check for empty	Return an empty instance
String	s == null	s._____	return
Array	arr == null	arr._____ == 0	return
ArrayList	list == null	list._____()	return
HashMap	map == null	map._____()	return

3. **Trace it (what happens and why?)**. For each line, write: **OK**, **compile error**, or **runtime error**, and name the error if applicable (e.g., syntax, null pointer, index out of bounds).

	Result	Why / Error name
`int[][] arr1 = null;`	_____	_____
`int[][] arr2 = new int[][];`	_____	_____
`int[][] arr3 = new int[0][0];`	_____	_____
`arr3[0][0] = 3;`	_____	_____
`System.out.println(arr1.length);`	_____	_____
`System.out.println(arr3.length);`	_____	_____

4. **Null + Methods**. If a variable is `null`, we CAN / CANNOT call a method on it.

5. **Review - Size of Objects**. Write the correct way to get the size/length:

```
String s = "Hello";                    s. ____________

int[] nums = {1,2,3,4,5};               nums.____________

ArrayList<String> list = new ArrayList<>();   list.____________

HashMap<String,Integer> map = new HashMap<>(); map.____________
```

6. **Access Modifier Order**. From most restrictive to least restrictive:

   _____

7. Identify the access modifier

   1) _____: available to any other class in the same package

   2) _____: can only be accessed within the declared class itself

   3) _____: accessible in the same package and by subclasses in other pkg

   4) _____: can be accessed from any class (=everywhere)

8. **Access Check (What can you access?)**

```
class Player {
    private int health = 100;
    int level = 1;
    protected int score = 50;
    public int coins = 10;
}
```

   A) Inside Player class. Which fields can be accessed?

   _____

   B) Different class, same package. Which fields can be accessed?

   _____

   C) Subclass in a different package. Which fields can be accessed?

   _____

D) Different package, NOT a subclass. Which fields can be accessed?

_____

E) Which modifier is best for encapsulating `health`?

_____

9. **Scope**

1) _____: anywhere in the object

2) _____: anywhere in the class

3) _____: within the method only

4) _____: within the block

10. **Scope in Action (What happens?)**

```
public class Player {
    int health = 100;

   public void setHealth(int health) {
        health = health;
    }
}
```

```
Player p = new Player();
p.setHealth(50);
System.out.println(p.health);
```

What is printed? _____

Fix the code:

```
  public  void setHealth(int health) {
        health = health;
```

11. **Before You Leave**

Write one question you still have about today's topic.

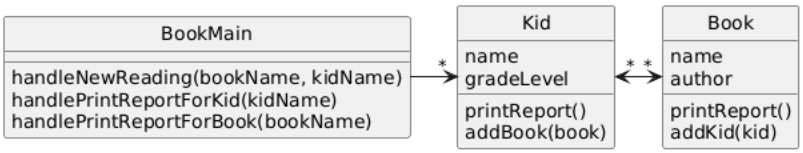
_____

Design Refactoring

Instructions (Day 1)

Book System: A website tracks books and the kids that read them. For each book the system stores the name and author. For each kid the system stores name and grade level. The teacher enters when a kid reads a particular book. It should be possible to print a report on a book that includes all kids who have read a particular book (with their grade level). It should be possible to print a report on a kid that includes the books (with authors) a particular kid has read.

Bad Design



Team Setup (Groups of 3). If only 2 - some roles must be combined

- Navigator: _____
- Driver: _____
- Recorder: _____

- Read the scenario and study the UML and provided code
- Copy the provided Google Doc template
- As a team, identify design problems in the system
- Propose a better design (you will implement it on Day 2)
- The Recorder must document your team's answers
- The Recorder must submit the link to the document and add team members in the submission.